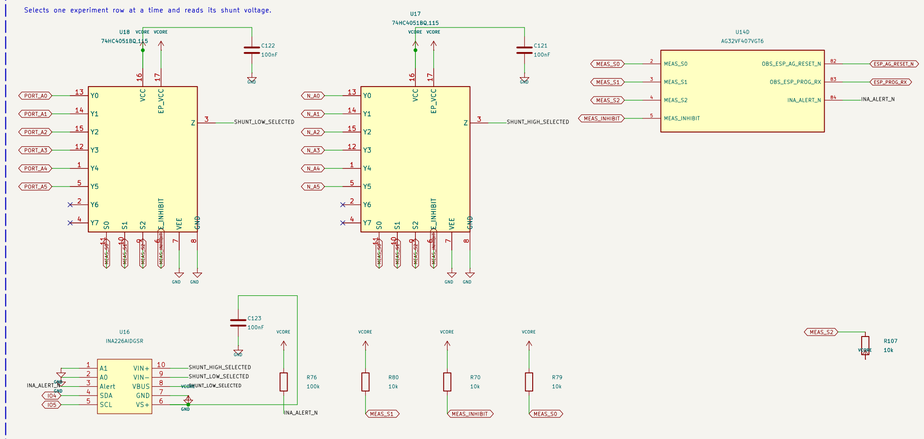
SPAGHETTILAB | EXPANSION — AG32 INSTRUMENT MODULE

Functional groups | local wires retained | power shown with symbols | cross-block links use named nets

U14A–U14G are units of ONE physical AG32. All measurements and FPGA features require firmware and prototype validation.

U14D + U16–U18 | SHARED CURRENT MONITOR

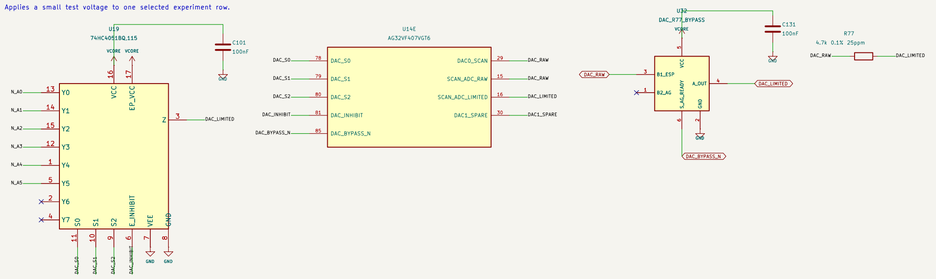
Selects one experiment row at a time and reads its shunt voltage.



Match both mux channel selections. Inhibit, select, settle, then start a fresh INA226 conversion. I04/I05 and mux commands are shared with ESP32.

U14E + U19 / U32 | ADJUSTABLE TEST SIGNAL

Applies a small test voltage to one selected experiment row.

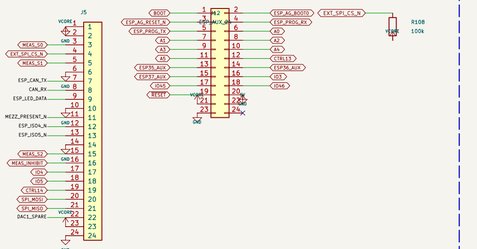




Internal ADC/DAC only. Scan with power MOSFETs OFF, other drivers Hi-Z and ESP bypass OPEN. Disable R77 bypass for limited stimulus. A return path is required; occupancy detection is not guaranteed.

J5 / J12 | EXTERNAL EXPANSION

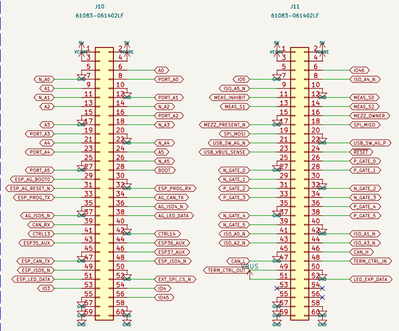
Brings ESP32 and supported AG32 signals to the flat cable and auxiliary header.



GPIO35/36/37 use carrier DNP links. Shared I2C/SPI and boot/strap pins are not unrestricted GPIO. All headers are 3.3 V logic.

J10 / J11 | CARRIER INTERFACE — 2 x 60 CONTACTS

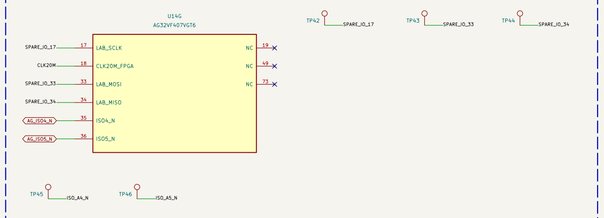
120 contacts: existing services + 22 future-module signals; J11 pins 53 / 54 / 56 remain reserved.



Future-module contacts are intentionally unused by this AG32 module. No new AG32 pins are assigned.

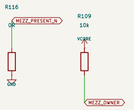
U14G | SPARE I/O AND ISOLATION CONTROL

Keeps spare controller connections and the last two port-enable signals accessible.



MODULE PRESENCE AND PRIORITY

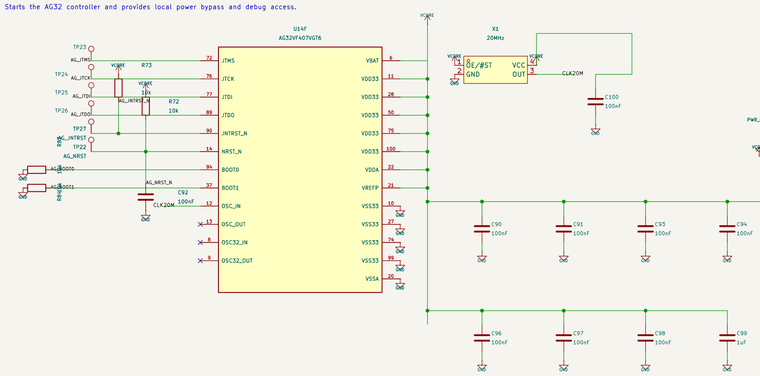
Reports the module presence and selects its shared services.



R109 = 10k pull-up claims services on insertion. R116 pulls MEZZ_PRESENT_N LOW. No automatic detection of a failed or unpowered fitted module.

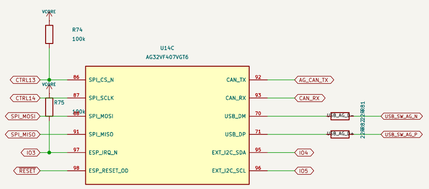
U14F + X1 | CLOCK, POWER AND RECOVERY

Starts the AG32 controller and provides local power bypass and debug access.



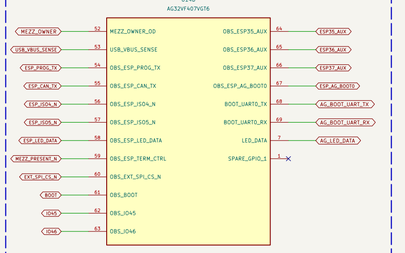
U14C | USB, CAN AND CONTROL LINKS

Exchanges messages with the computer, carrier and optional ESP32.



U14B | OWNERSHIP AND GPIO OBSERVATION

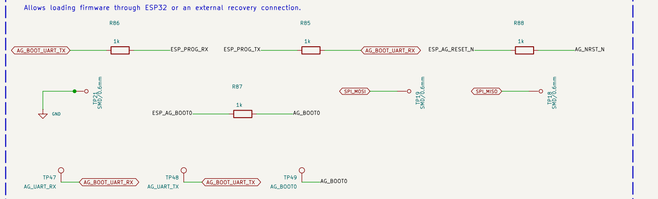
Manages the module role and observes additional ESP32 signals.



AG52: release or drive LOW in open-drain; do not drive HIGH. OBS inputs stay Hi-Z. FPGA routing and protocol bridges still require toolchain validation.

ESP32 TO AG32 | PROGRAMMING AND RECOVERY

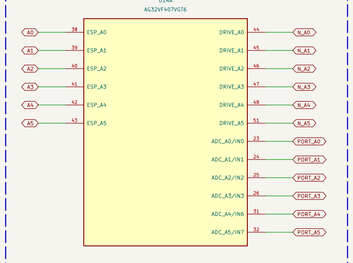
Allows loading firmware through ESP32 or an external recovery connection.



ESP TX/RX -> AG UART RX/TX. Boot and reset access are preserved. Recovery requires ESP pins released. Firmware and bootloader sequence are not implemented by this schematic.

U14A | SIX-PORT OBSERVATION

Observes the experiment while leaving the direct ESP32 path available.



A0–A5 = ESP side; N_A0–A5 = driver side; PORT_A0–A5 = external side after shunt. Inline FPGA requires carrier bypass switches OPEN and validated HDL.

Functional clusters; wiring preserved. Not production-qualified.

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Title: SpaghettiLAB — Expansion	
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